12.6 Case Study: The IEEE 802.11a Wireless LAN Standard

The IEEE 802.11a wireless LAN standard, which occupies 300 MHz of bandwidth in the 5-GHz unlicensed band, is based on OFDM [6]. The IEEE 802.11g standard is virtually identical to the 802.11a standard in its link layer design, but it operates in the smaller and more crowded 2.4-GHz unlicensed ISM band [7]. In this section we study the properties of this OFDM design and discuss some of the design choices.

In 802.11a, $N = 64$ subcarriers are generated, although only 48 are actually used for data transmission: the outer 12 are zeroed in order to reduce adjacent channel interference, and 4 are used as pilot symbols for channel estimation. The cyclic prefix consists of $\mu = 16$ samples, so the total number of samples associated with each OFDM symbol, including both data samples and the cyclic prefix, is 80. Coded bits are packetized, and the transmitter gets periodic feedback from the receiver about the packet error rate, which it uses to pick an appropriate error correction code and modulation technique. The same code and modulation must be used for *all* the subcarriers at any given time. The error correction code is a convolutional code with one of three possible code rates: $r = 1/2$, $2/3$, or $3/4$. The modulation types that can be used on the subchannels are BPSK, QPSK, 16-QAM, or 64-QAM.

The total system bandwidth of 300 MHz is divided into 20-MHz channels that can be assigned to different users. Since the channel bandwidth B (and sampling rate $1/T_s$) is 20 MHz and since there are 64 subcarriers evenly spaced over that bandwidth, the subcarrier bandwidth is:

$$B_N = \frac{20 \text{ MHz}}{64} = 312.5 \text{ kHz}.$$

Since $\mu = 16$ and $1/T_s = 20$ MHz, the maximum delay spread for which ISI is removed is

$$T_m < \mu T_s = \frac{16}{20 \text{ MHz}} = 0.8\ \mu\text{s},$$

which corresponds to delay spread in an indoor environment. Including both the OFDM symbol and cyclic prefix, there are $80 = 64 + 16$ samples per OFDM symbol time, so the symbol time per subchannel is

$$T_N = 80T_s = \frac{80}{20 \cdot 10^6} = 4\ \mu\text{s}.$$

The data rate per subchannel is $\log_2 M/T_N$. Thus, the minimum data rate for this system – corresponding to BPSK (1 bit/symbol), an $r = 1/2$ code, and taking into account that only 48 subcarriers actually carry usable data – is given by

$$R_{\min} = 48 \text{ subcarriers} \cdot \frac{1/2 \text{ bit}}{1 \text{ coded bit}} \cdot \frac{1 \text{ coded bit}}{\text{subcarrier symbol}} \cdot \frac{1 \text{ subcarrier symbol}}{4 \cdot 10^{-6} \text{ seconds}}$$
$$= 6 \text{ Mbps.} \tag{12.49}$$

The maximum data rate that can be transmitted is

$$R_{\max} = 48 \text{ subcarriers} \cdot \frac{3/4 \text{ bit}}{1 \text{ coded bit}} \cdot \frac{6 \text{ coded bits}}{\text{subcarrier symbol}} \cdot \frac{1 \text{ subcarrier symbol}}{4 \cdot 10^{-6} \text{ seconds}}$$
$$= 54 \text{ Mbps.} \tag{12.50}$$

Naturally, a wide range of data rates between these two extremes is possible.

EXAMPLE 12.6: Find the data rate of an 802.11a system assuming 16-QAM modulation and rate-$2/3$ coding.

Solution: With 16-QAM modulation, each subcarrier transmits $\log_2(16) = 4$ coded bits per subcarrier symbol and there are a total of 48 subcarriers used for data transmission. With a rate-$2/3$ code, each coded bit relays $2/3$ of an information bit per T_N seconds. Thus, the data rate is given by

$$R_{\max} = 48 \text{ subcarriers} \cdot \frac{2/3 \text{ bit}}{1 \text{ coded bit}} \cdot \frac{4 \text{ coded bits}}{\text{subcarrier symbol}} \cdot \frac{1 \text{ subcarrier symbol}}{4 \cdot 10^{-6} \text{ seconds}}$$
$$= 32 \text{ Mbps.} \tag{12.51}$$

Reference: Andrea Goldsmith, “Wireless Communications”, Cambridge University Press, 2005